

Data Fusion for Nowcasting

Roshni Sahoo

`r.sahoo@columbia.edu`

Columbia University

Abstract

Policy makers often need to make decisions that perform well over a target population in real-time. We study decision making given representative but lagged data from a target population and real-time but potentially biased data drawn from a survey population that differs from the target population. When the selection mechanism into the survey population is stationary, we demonstrate that the optimal decision rule is the solution to a weighted risk minimization problem on the survey data, where the weights are time-invariant and identified as the solution to an augmented convex optimization problem on data pooled from the target and survey populations across time.

1 Introduction

Decision makers aim to deploy policies that perform well under the current target population. However, the target population can evolve in the time between representative data collection for informing policy occurs and when the policy is deployed. For example, [Aiken et al. \[2023\]](#) observe that proxy-means tests, or policies for determining eligibility for social protection programs in lower- and middle-income countries, are often trained on in-person household survey data collected years prior to a policy deployment, and in an evaluation of such policies over time, they find that the policies can suffer from increasing rates of inclusion and exclusion errors as the time from the survey increases.

During the COVID-19 pandemic, the World Bank Living Standards Measurement Study launched a series of telephone surveys to facilitate rapid data collection. Telephone surveys are potentially a powerful technique for facilitating rapid and large-scale data collection, at a fraction of the cost of in-person surveys. However, they may be biased relative to the target population because the people who opt in to answering a telephone survey may not be representative of the target population. As a result, a decision rule that is based solely on the data from the telephone survey alone may also perform poorly when deployed to the current target population.

In this work, we consider a decision maker who aims to deploy a decision rule in the current target population given access to two data sources, lagged data from the target population and real-time data from the survey population. We envision that the lagged data from the target population is collected through a in-person LSMS-style survey, while the real-time data from the survey population is collected from a telephone or web survey. We demonstrate that when the selection mechanism from target population into the survey population is stationary over time, we can recover the optimal policy under the target population in the current time-step.

1.1 Related Work

Our work is a contribution to the burgeoning literature on data fusion, which seeks to combine data sources with different properties for prediction and decision making. Guan et al. [2025] consider combining individual-level samples from a survey population and aggregate-level information from the target population.

A key component of our approach is that our weighting function can be expressed as the solution to a convex loss minimization problem under a distribution that pools both target and survey data sources over time. This technique is motivated by previous works that reduce density ratio estimation to a binary classification problem [Bickel et al., 2009, Menon and Ong, 2016, Sugiyama et al., 2008]. In particular, our objective in (4) is equivalent to a transformation of the KLIEP classification objective Menon and Ong [2016] that targets the log density ratio when conditioned on a single time-step t and minimized with respect to α . Nevertheless, in this work, we estimate a weighting function that is common to the factorizations of the density ratio between multiple pairs of target and survey distributions, rather than the density ratio between a single pair of distributions. This technique is enabled by the observation that the density ratio at any time step can be factored into a time-dependent nuisance component and an outcome-dependent function under stationary selection.

Our notion of stationary selection is most closely related to the Cox proportional hazard model from survival analysis [Cox, 1972]. The Cox proportional hazards model assumes that the hazard rate factors into a product of a time-dependent function and a covariate-dependent function; equivalently, the ratio between the hazard rates of units with different covariates is constant over time. However, stationarity yields different implications in our work compared to survival analysis literature. In survival analysis, the estimand of interest is the hazard rate itself, so stationarity permits simplified modeling of the hazard rate but the estimand still depends on the time-dependent quantity. In our work, our estimand of interest is the target population risk, and under stationarity, the time-dependent factor completely cancels out from the estimand.

The problem of learning and estimation under nonignorable selection has also received attention in statistics [Franks et al., 2020, Kim and Yu, 2011, Uehara et al., 2023]. A central challenge is that the target estimand is not identifiable from the survey population alone, so prior works often posit a selection model and assume access to the additional data to identify the target parameters. One family of approaches assumes access to a nonresponse instrumental variable to identify the target parameter [Kim and Yu, 2011, Uehara et al., 2023]. A second family of approaches [Magder, 2003, Miao et al., 2025] attains identification through stationarity assumptions. An interesting finding across these works is that they model selection bias or relate the distributions of respondents and non-respondents through an exponential tilting model, but in our work, the exponential tilting arises as a consequence of our stationarity assumption.

2 Statistical Setup

Let S_t, P_t denote the survey and target population distributions respectively at time-step t . Let P_t be a distribution over a covariate $X \in \mathcal{X}$, outcome $Y \in \mathcal{Y}$, and a binary response indicator $R \in \{0, 1\}$. Let S_t be the $X, Y \mid R = 1$ marginal of P_t . We have access to S_t for $t = 1, \dots, M$ and P_t for $t = 1, \dots, m$, where $m < M$. For simplicity, we will assume that the covariate distribution of the survey and population distributions are the same.

We aim to learn a decision rule h that performs well under the target population P_t

$$\min_{h: \mathcal{X} \rightarrow \mathbb{R}} \mathbb{E}_{P_t} [L(h(X), Y)]. \quad (1)$$

We define relative probability of sample selection between two individuals with outcome y and y' at time-step t as

$$w(x, y, y'; t) = \frac{\mathbb{P}_{P_t}[R = 1 \mid X = x, Y = y]}{\mathbb{P}_{P_t}[R = 1 \mid X = x, Y = y']}. \quad (2)$$

Assumption 1 (Stationary Selection). The function $w(x, y, y'; t)$ is constant in t .

Lemma 1. *The population and survey conditional distributions are related via a time-invariant exponential tilting, i.e. there exists some $\theta : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ such that*

$$dP_{t,Y|X}(y) = dS_{t,Y|X}(y) \cdot \exp(\theta(x, y) - A_{t,x}(\theta)),$$

where $A_{t,x}(\theta) = \log \left(\int \exp(\theta(x, y)) \cdot dS_{t,Y|X=x}(y) dy \right)$, if and only if Assumption 1 holds.

This result implies that

$$\mathbb{E}_{P_t}[L(h(X), Y)] = \mathbb{E}_{S_{t,X}} \left[\mathbb{E}_{S_{t,Y|X}} [\exp(\theta(X, Y)) \cdot L(h(X), Y) \mid X] / \mathbb{E}_{S_{t,Y|X}} [\exp(\theta(X, Y)) \mid X] \right].$$

3 Algorithm

Define a pooled distribution F over paired data from S_t, P_t for $t = 1, 2, \dots, m$. Let T be a random variable that denotes the time step and let Z be a pseudolabel that captures whether a sample is drawn from the survey or target population. The distribution F is a distribution over (X, Y, Z, T) , where $Z \sim \text{Bernoulli}(1/2)$ and $T \sim \text{Uniform}([m])$, $F_{X,Y|Z=1,T=t} = P_t$, and $F_{X,Y|Z=0,T=t} = S_t$. Let Θ be the space of square-integrable functions with measurable functions with respect to F_Y , i.e. $\Theta := L^2(F_{X,Y}, \mathcal{X} \times \mathcal{Y})$. Let $\mathcal{A} \subset L^2(F_{X,T}, \mathcal{X} \times \mathbb{R})$ with last coordinate equal to zero, i.e. $\mathcal{A} = \{\alpha \in \mathbb{R}^m \mid \alpha(\cdot, m) = 0\}$. Define a loss function

$$L(\theta, a, z) := (1 - z) \cdot \exp(\theta + a) - z \cdot (\theta + a). \quad (3)$$

We consider solving the augmented convex optimization problem

$$\{\tilde{\theta}(\cdot), \tilde{\alpha}(\cdot)\} \in \arg \min_{(\theta, \alpha) \in \Theta \times \mathcal{A}} \{\mathbb{E}_F[L(\theta(X, Y), \alpha(X, T), Z)] : \alpha(\cdot, m) = 0\}. \quad (4)$$

It is straightforward to see that given access to the density ratio between the survey and population distribution at the current time-step, one can reweight data from the survey population

Lemma 2. *The solution to (4), denoted by $(\tilde{\theta}, \tilde{\alpha})$, exists and is unique. If Assumption 1 holds, h that solves (1) also solves*

$$\min_{h: \mathcal{X} \rightarrow \mathbb{R}} \mathbb{E}_{S_t} [\exp(\tilde{\theta}(X, Y)) \cdot L(h(X), Y)].$$

Remark 1. A corollary of this result is that if we aim to learn the optimal decision rule under P_t from an unrestricted model class and Assumption 1 holds, then it suffices to minimize a reweighted risk, where the weights are determined by the density ratio between any pair of target and survey joint distributions, i.e.

$$\min_{h: \mathcal{X} \rightarrow \mathbb{R}} \mathbb{E}_{S_t} \left[\frac{dP_j(X, Y)}{dS_j(X, Y)} \cdot L(h(X), Y) \right], \quad (5)$$

where $j \in [m]$. Note that the above objective is not equal to the current target risk $\mathbb{E}_{P_t}[L(h(X), Y)]$. Nevertheless, the optimal decision rule that solves (5) also minimizes the current target risk.

Suppose that $\mathcal{H}$ is a restricted policy class. Then, the decision rule from this class that minimizes the target risk is given by

$$\min_{h \in \mathcal{H}} \mathbb{E}_{P_t} [L(h(X), Y)]. \quad (6)$$

If Assumption 1 holds, then h that solves (6) also solves

$$\min_{h \in \mathcal{H}} \mathbb{E}_{S_t} \left[r_t(X) \cdot \frac{\exp(\tilde{\theta}(X, Y))}{A_{t,X}(\tilde{\theta})} \cdot L(h(X), Y) \right],$$

where $A_{t,x}(\theta) := \mathbb{E}_{S_t} [\exp(\theta(X, Y)) \mid X = x]$ and $r_t(x) := dP_{t,X}(x)/dS_{t,X}(x)$.

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